A. CQXYM Count Permutations

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: combinatorics, math, number theory | https://codeforces.com/problemset/problem/1581/A

CQXYM is counting permutations length of $2n$.

A permutation is an array consisting of n distinct integers from 1 to n in arbitrary order. For example, [2,3,1,5,4] is a permutation, but [1,2,2] is not a permutation (2 appears twice in the array) and [1,3,4] is also not a permutation (n=3 but there is 4 in the array).

A permutation p(length of $2n$) will be counted only if the number of i satisfying p_i lt p_{i+1} is no less than n. For example:

- Permutation [1, 2, 3, 4] will count, because the number of such i that p_i lt p_{i+1} equals 3 (i = 1, i = 2, i = 3).
- Permutation [3, 2, 1, 4] won't count, because the number of such i that p_i lt p_{i+1} equals 1 (i = 3).

CQXYM wants you to help him to count the number of such permutations modulo 1000000007 (10^9+7).

In addition, modulo operation is to get the remainder. For example:

- 7 mod 3=1, because 7 = 3 · 2 + 1,
- 15 mod 4=3, because 15 = 4 · 3 + 3.

Input

The input consists of multiple test cases.

The first line contains an integer t ($t \geq 1$) — the number of test cases. The description of the test cases follows.

Only one line of each test case contains an integer n($1 \leq n \leq 10^5$).

It is guaranteed that the sum of n over all test cases does not exceed 10^5

Output

For each test case, print the answer in a single line.

Example

Input
4 1 2 9 91234

Output
1 12 830455698 890287984

Note

n=1, there is only one permutation that satisfies the condition: [1,2].

In permutation [1,2], p_1 lt p_2, and there is one i=1 satisfy the condition. Since 1 ≥ n, this permutation should be counted. In permutation [2,1], p_1 gt p_2. Because 0 lt n, this permutation should not be counted.

n=2, there are 12 permutations: [1,2,3,4],[1,2,4,3],[1,3,2,4],[1,3,4,2],[1,4,2,3],[2,1,3,4],[2,3,1,4],[2,3,4,1],[2,4,1,3],[3,1,2,4],[3,4,1,2],[4,1,2,3].